

Chapter 3: Perfect-Information Games (aka Game Trees)

ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

9/24

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Everything So Far Happened at Once

In Chapter 2, players chose **simultaneously**, seeing nothing beforehand.

But most interesting situations aren't like that:

- chess
- an offer and a counteroffer
- a firm entering a market and the incumbent responding
- calling a play after seeing the defense line up

Now: players move **in turn**, and everyone sees the whole history.

Moblab: “Centipede Game”

- MobLab time!
- Two players building a firm over time
- As long as players invest, firm doubles in value each day
- Each “turn” one player chooses:
 - **Invest:** keep the firm going.
 - **Cash Out:** sell the firm, take 80% of its value. Game over.
 - Note: If the other person cashes out, you only get 20%!!
- Alternating moves (P1, P2, P1, P2...)
- If value gets to 80, P1 must Cash Out, giving (64, 14)

Feel free to talk! And play it the way you'd actually play it.

How Far Did You Get?

Most frequent # of Turns:

3

Groups that Cashed Out in Turn 1:

Groups that made it to the end:

Trees

Perfect Information

In a **dynamic game** (or **extensive form**) players move in sequence.

This chapter covers the case of **perfect information**:

whenever it is a player's turn to move,
she knows every move made so far

- **Perfect Information Games:**

- Chess, checkers, Connect Four, bargaining, sequential legislative agenda/votes

- **Imperfect Information Games:**

- Stratego, Battleship, Mastermind, patent races, dynamic attacker/defender games
- Any game with a hidden move by “Nature” (see Chapter 4)
 - Poker, auctions, bidding wars, diplomatic negotiations w/ private info

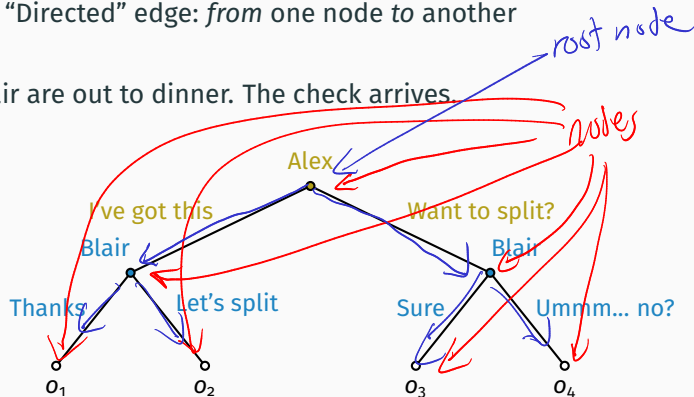
Rooted Directed Trees

We represent these games using rooted directed trees.

A **rooted directed tree** is a set of **nodes** joined by directed **edges**.

“Directed” edge: *from* one node to another

Ex: Alex and Blair are out to dinner. The check arrives.



Rooted Directed Trees

A **rooted directed tree** is a set of **nodes** joined by directed **edges**

- The **root** node has no edge leading into it
- Every other node has **exactly one** edge leading into it
 - Branches out, like a tree
 - Contains *no cycles/loops*
- **Terminal nodes** have no edges going out
 - They are the various game endings
 - In a game, each has an outcome o attached to it
- there is a **unique path** from the root to any other node

starting point



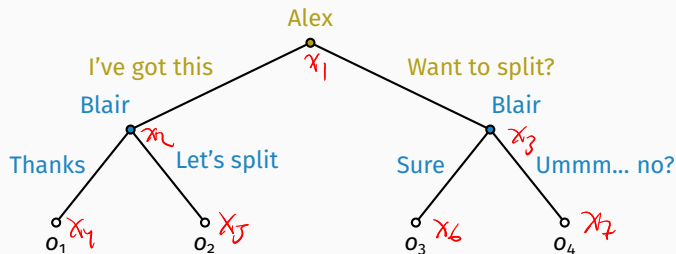
Set of **terminal nodes**: Z . All others: "**Decision nodes**" D . Then define the set of **all nodes**:

$$X = D \cup Z$$

all nodes $\rightarrow$ X $\leftarrow$ decision $\leftarrow$ terminal

Each node $x \in X$ represents a "history" of the game so far

Back to the Example



$$D = \{x_1, x_2, x_3\}$$

$$Z = \{x_4, x_5, x_6, x_7\}$$

$$X = D \cup Z$$

Definition: Extensive Form With Perfect Information

A **finite extensive form with perfect information** consists of:

- A finite rooted directed tree *tree*
- A set of players $I = \{1, \dots, n\}$
 - Each decision node is assigned to one player *who gets to move at every decision node*
- A set of actions A
 - Each edge is assigned one action
 - Two edges from the same node $\Rightarrow$ different actions *available options at every node*
- A set of outcomes O
 - Each terminal node is assigned one outcome *endings*

Note: no **preferences** yet! This is a frame, not a game. Outcomes only.

Dissolving the Partnership

Nocterra Brewing Company shut down.

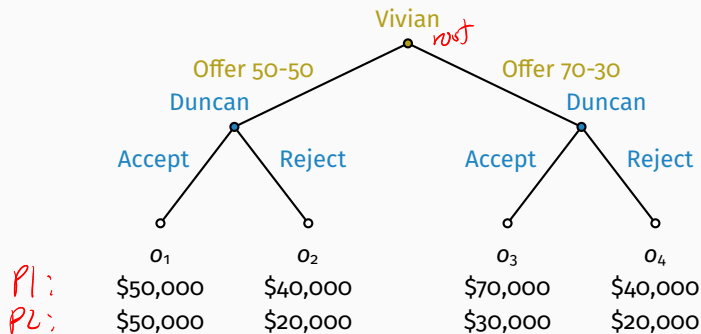
Owners: Vivian and Duncan

Suppose remaining assets are valued at \$100,000. How to split it?

- The charter says the senior partner, **Vivian**, proposes a split
- **Duncan** can Accept, or Reject and take it to court
- In court: each side pays \$20,000 in legal fees, and the verdict will 60% to Vivian, 40% to Duncan (no uncertainty)
- Vivian is allowed only two possible offers: **50-50** or **70-30**

So rejecting yields Vivian $\$60,000 - \$20,000 = \$40,000$
and Duncan $\$40,000 - \$20,000 = \$20,000$.

The Frame



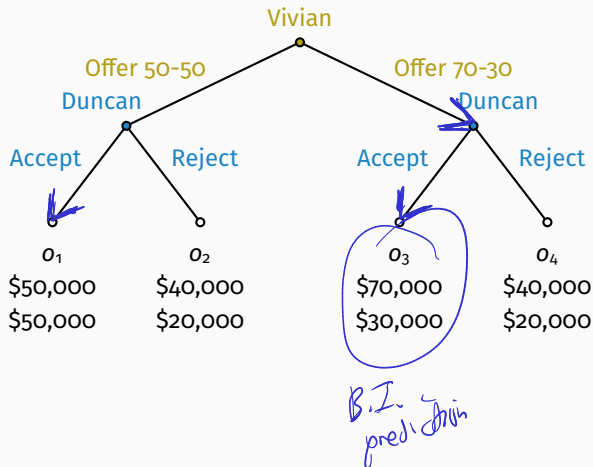
Top number is Vivian's (P1), bottom number is Duncan's (P2).

For now let's assume *selfish & greedy* preferences.

Backward-Induction Reasoning

Let's build up a new solution concept

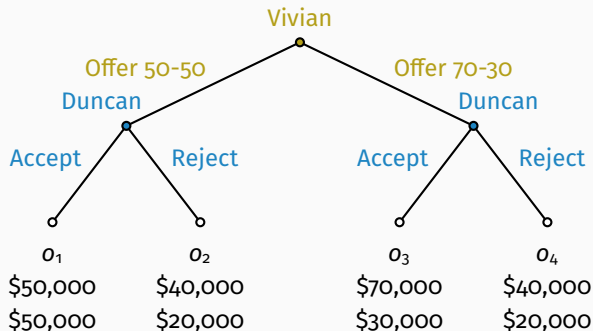
Backwards induction: Start at the *end* and work back toward the root.



Backward-Induction Reasoning

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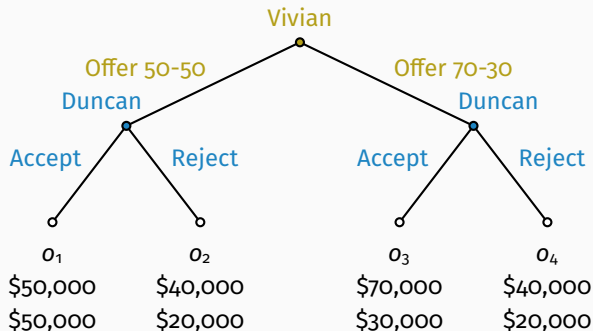


- Duncan accepts at both nodes

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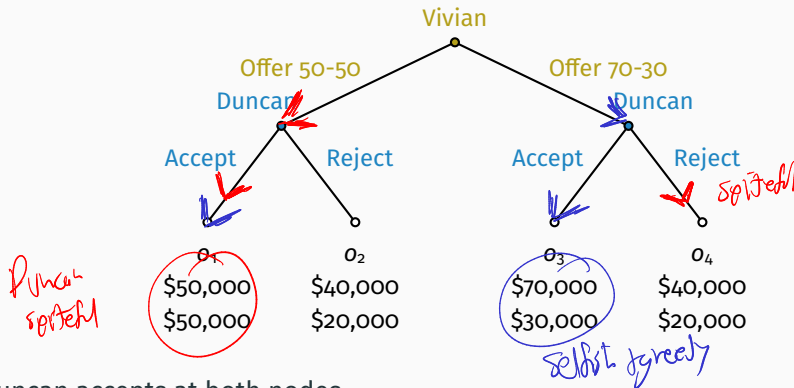


- Duncan accepts at both nodes
- Knowing this, Vivian offers 70-30

Backward-Induction Reasoning

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- Duncan accepts at both nodes
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Another solution concept built on “common knowledge of rationality”

Non-Selfish Preferences?

What if they are *not* selfish & greedy??

Suppose instead Duncan thinks they worked just as hard as Vivian, and that the only fair split is 50-50. They may feel strongly enough that they would give up \$10,000 to refuse an unfair offer.

That is: $o_4 \succ_{\text{Duncan}} o_3$. 

This will change the predicted solution:

Duncan Accepts 50-50 but *Rejects* 70-30. So now Vivian offers 50-50.

A tree with *outcomes* at the terminal nodes is a **frame**.

Replace them with the players' *utility values* and you have a **game**.

Adding Preferences: An Example

Suppose Vivian is selfish and greedy:

$$\overset{70}{O_3} \succ_1 \overset{50}{O_1} \succ_1 \overset{40}{O_2} \sim_1 \overset{40}{O_4}$$

Duncan cares about fairness:

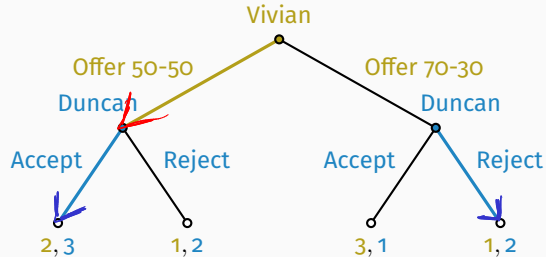
$$\overset{50}{O_1} \succ_2 \overset{20}{O_2} \sim_2 \overset{20}{O_4} \succ_2 \overset{30}{O_3}$$

	O_1	O_2	O_3	O_4
U_1 (Vivian)	2	1	3	1
U_2 (Duncan)	3	2	1	2

← arbitrary
any utility
values
that respect

Now replace each outcome with its payoff pair to get the actual **game**

The Game

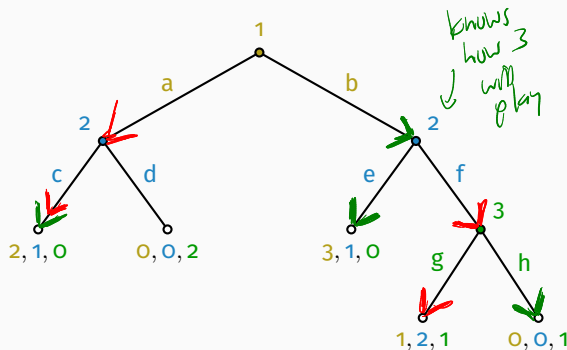


Colored edges show the backward-induction action choices.
Vivian now anticipates 70-30 will be rejected, offers 50-50.

Backward Induction: How to Handle Ties

With ties, there can be multiple backwards induction predictions:

Red:
B.I. prediction
assuming g
a, c



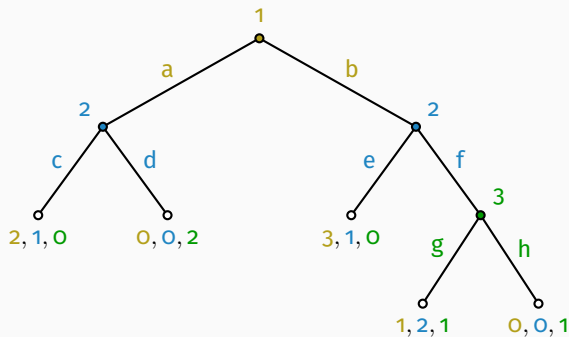
Green:
B.I. prediction
assuming h
b, e

Payoffs in order: Player 1, Player 2, Player 3.

Strategies

Actions vs. Strategies

A **very** important distinction: actions vs. strategies



Actions:

$$A_1 = \{a, b\}$$

$$A_2 = \{c, d, e, f\}$$

$$A_3 = \{g, h\}$$

Strategies: "complete plans of actions"

$$S_1 = \{a, b\}$$

$$S_2 = \{(c, e), (c, f), (d, e), (d, f)\}$$

$$S_3 = \{g, h\}$$

A Strategy Is a Complete Plan

Definition

A **strategy** for a player in a perfect-information game is a list of *actions*, **one for each decision node of that player**.

Not “what I will do” but “what I *would* do everywhere I might move.”

In the previous game, Player 2 has two decision nodes, so a strategy is a pair. Before the game starts she doesn't know whether Player 1 will pick *a* or *b*, so she must plan for both:

(c, e) (c, f) (d, e) (d, f)

Four strategies, not two.

Counting Strategies

Suppose Player 1 has three decision nodes in some game:

- at the first, three choices a_1, a_2, a_3
- at the second, two choices b_1, b_2
- at the third, four choices c_1, c_2, c_3, c_4



A strategy fills in three blanks:

$$\left(\underbrace{\hspace{2cm}}, \underbrace{\hspace{2cm}}, \underbrace{\hspace{2cm}} \right)$$

one of a_1, a_2, a_3 one of b_1, b_2 one of $c_1, \dots, c_4$

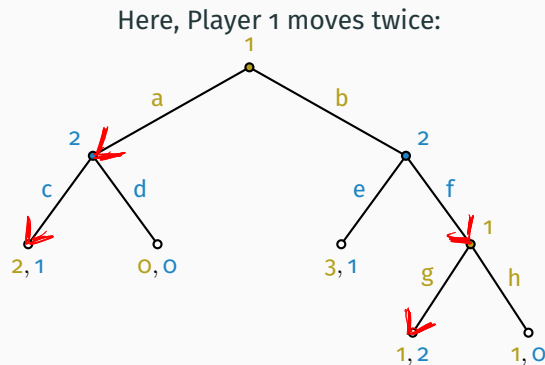
So her strategy set is a Cartesian product of the action sets she faces:

$$S_1 = \{a_1, a_2, a_3\} \times \{b_1, b_2\} \times \{c_1, c_2, c_3, c_4\}$$

$$3 \times 2 \times 4 = 24 \text{ strategies}$$

Strategy sets get large fast! Because they're a complete plan.

Don't Throw Out Impossible Strategies



There are 2 backwards-induction predictions:

But strategy (a, g) is impossible! Playing a makes g irrelevant.

But let's keep it. Player 2's choice of f depended on g . Backwards induction requires a specification of how you play *everywhere*.

Other Justifications

Other ways to justify specifying (a, g) (not just (a)):

1. **Mistakes.** She's cautious enough to plan for the possibility that she intends a but ends up playing b .
2. **Delegation.** A strategy is a set of instructions handed to someone playing on her behalf, who needs to know what to do everywhere.
3. **Beliefs.** A strategy isn't really Player 1's plan at all, but Player 2's *belief* about what Player 1 would do.

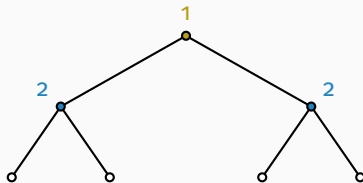
But the main reason: **Backwards induction needs it**

On Path and Off Path

A strategy profile traces one path from the root to a terminal node.

- Nodes on that path are **on the path of play** (“on path”)
- Every other node is **off the path of play** (“off path”)

A strategy must still say what to do off path. Even if certain actions aren't on-path, they make the on-path actions rational.



From Tree to Table

Every strategy profile determines a unique path, hence a unique terminal node, hence a unique payoff vector.

So every perfect-information game has an **associated strategic form**.

		Player 2			
		ce	cf	de	df
Player 1	ag	2, 1	2, 1	0, 0	0, 0
	ah	2, 1	2, 1	0, 0	0, 0
	bg	3, 1	1, 2	3, 1	1, 2
	bh	3, 1	1, 0	3, 1	1, 0

The top two rows are identical. That's the redundancy showing up.

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So every perfect-information game has an **associated strategic form**.

		Player 2			
		ce	cf	de	df
Player 1	ag	2, <u>1</u>	<u>2</u> , <u>1</u>	0, 0	0, 0
	ah	2, <u>1</u>	<u>2</u> , <u>1</u>	0, 0	0, 0
	bg	<u>3</u> , 1	1, <u>2</u>	<u>3</u> , 1	<u>1</u> , <u>2</u>
	bh	<u>3</u> , 1	1, 0	<u>3</u> , <u>1</u>	<u>1</u> , 0

The top two rows are identical. That's the redundancy showing up.

Five Equilibria, Two Solutions

That game has **five** Nash equilibria:

(ag, cf) (ah, cf) (bg, df) (bh, ce) (bh, de)

It has **two** backward-induction solutions:

- $((a, g), (c, f))$ (because g and h tie for Player 1)
- $((b, h), (c, e))$

Both are Nash equilibria.

The other three are not backward-induction solutions.

That gap is the whole point of the next section.

Solution vs. Outcome

A backward-induction **solution** is a *strategy profile*. It describes behavior everywhere, including at nodes that never get reached.

A backward-induction **outcome** is the *sequence of moves actually made*.

Solution $((a, g), (c, f))$ outcome ac , payoffs $(2, 1)$

Solution $((b, h), (c, e))$ outcome be , payoffs $(3, 1)$

Keep the distinction straight. Exam questions live here.

Backward Induction & Nash

The Theorem

Theorem. Every backward-induction solution of a perfect-information game is a Nash equilibrium of the associated strategic form.

Suppose player i deviates.

- If she only changes what she does off the path, the play doesn't change, so her payoff doesn't either
- If she changes what she does at a node *on* the path, the algorithm already checked that alternative and rejected it

So backward induction is a **refinement** of Nash equilibrium: fewer solutions, all of them Nash.

Which Nash Equilibria Get Cut?

Usually the ones resting on an **incredible threat**.

The Entry Game

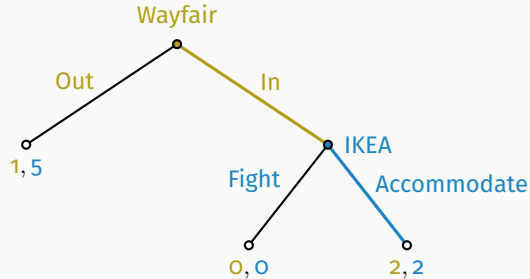
IKEA has been the only big furniture store at Polaris for years, earning \$5 million. Wayfair is deciding whether to open across I-71.

- If Wayfair stays **out**, it earns \$1 million elsewhere
- If Wayfair enters, IKEA can **fight**, a price war leaving both at zero
- Or IKEA can **accommodate** and share, and both make \$2 million

Both firms are selfish and greedy.

IKEA announces: “open here and we will bury you on price.”
Should Wayfair believe it?

The Entry Game



First number Wayfair, second number IKEA.

Backward induction: faced with entry, IKEA compares 0 to 2. It accommodates. Knowing that, Wayfair enters.

But There Are Two Nash Equilibria

		IKEA	
		Fight	Accommodate
Wayfair	In	0, 0	<u>2</u> , <u>2</u>
	Out	<u>1</u> , <u>5</u>	1, <u>5</u>

- (In, Accommodate), the backward-induction solution
- (Out, Fight), also a Nash equilibrium!

It survives only because Wayfair never enters, so the threat is untested.

It is an **incredible threat**: if entry actually happened, IKEA would not want to carry it out.

Selten's Chain-Store Game

IKEA also has a store in Cincinnati, and Wayfair could expand there next. So the entry decision happens **one city at a time**, with everyone watching what happened in the earlier one.

IKEA makes the argument in CBus:

Fighting you costs me \$2 million today. But Cincy is watching. If they see me fight, they may stay out, and then I make $\$(0+5) = \5 million across the two cities. If I accommodate you here, they enter there too, and I make $\$(2+2) = \4 million. So it is in my interest to fight. Stay out.

Sounds right. Does backward induction agree?

The Reply

No, and the reason is where backward induction starts.

- Whatever happens in CBus, Cincy is the **last** interaction
- Nobody is watching after that, so no reputation is left to build
- So Cincy is just the one-shot entry game, and IKEA accommodates
- Which means Cincy goes the same way *regardless* of CBus
- So fighting in CBus buys IKEA nothing: $\$(0 + 2) = \2 million against $\$(2 + 2) = \4 million

Entry happens in both cities. IKEA accommodates both.

What That Tells Us

The reputation argument is intuitive and backward induction rejects it.

- That's not a flaw in the arithmetic. The unraveling is airtight
- The reputation story needs **uncertainty**: the Cincy entrant must be unsure what kind of opponent IKEA is
- In a perfect-information game, uncertainty is ruled out by definition

We'll come back to this with better tools. For now, note the limitation.

Backward Induction and IDWDS

Backward induction resembles deleting dominated choices at each node.

Applying IDWDS to the strategic form of a perfect-information game gives profiles containing **at least one** backward-induction solution.

But that's all you can say in general. The IDWDS output:

- may also contain profiles that are **not** backward-induction solutions
- may **miss** some backward-induction solutions

So they're related, not the same. Don't use one to compute the other.

Win, Lose, or Draw

Win-Lose Games

Two players, two outcomes: Player 1 wins (W_1) or Player 2 wins (W_2).

Player 1 strictly prefers W_1 ; Player 2 strictly prefers W_2 . Use payoffs $(1, 0)$ and $(0, 1)$.

Theorem. In every finite two-player win-lose game with perfect information, **one of the two players has a winning strategy.**

A winning strategy guarantees a win *no matter what the opponent does*.

Note what it doesn't say: *which* player, or what the strategy is.

Let's Play

Two players take turns choosing a number from $\{1, 2, \dots, 10\}$.

Running total starts at 0. **First to reach 100 or more wins.**

Player 1 moves first.

Volunteer? I'll go second.

Why I Was Always Going to Lose That

The tree is far too big to draw: 10,000 nodes just for the first four moves. But we can reason backward anyway.

Call a total a **losing position** if whoever has to move from it cannot win.

- **89** is a losing position: from 89 you can reach only 90, ..., 99, and from any of those I can hit exactly 100
- **78** is the next one down: from 78 you must reach 79, ..., 88, and from any of those I can get to 89
- Keep going: 89, 78, 67, 56, 45, 34, 23, 12, **1**

Player 1's Winning Strategy

Start with **1**.

Then every turn, choose **$11 - n$** , where n is what your opponent just chose.

Each pair of moves adds exactly 11, so the total marches 1, 12, 23, $\dots$, 89, 100.

Notice this is a genuine *strategy*: it says what to do at every position Player 1 could face, not just the ones on the winning path.

Adding a Draw

Now three outcomes: W_1 , W_2 , and a draw D , with

$$W_1 \succ_1 D \succ_1 W_2 \quad \text{and} \quad W_2 \succ_2 D \succ_2 W_1$$

Theorem. Every such game falls into exactly one of three categories:

1. Player 1 has a strategy guaranteeing W_1
2. Player 2 has a strategy guaranteeing W_2
3. Player 1 can guarantee *at least* a draw, and so can Player 2, so if both play well it's a draw

The proof is the same backward-induction argument, with three possible markings instead of two.

Which Category?

- **Tic-Tac-Toe:** category 3. Solved: perfect play draws.
- **Checkers:** category 3. Solved in 2007, after 18 years of work.
- **Chess:** _____

The theorem says chess is in *one* of the three categories. It does not say which, and nobody knows.

Existence proofs are like that.

Back to the Centipede

Apply backward induction to the game you played:

- At the **last** decision, taking beats passing. So take.
- At the second-to-last, you know the other player will take next, so passing gets you the smaller share. So take.
- Keep unraveling ... all the way to the first move.

Prediction: the very first player takes immediately.

Our numbers said: _____

So Who's Wrong?

Three honest possibilities:

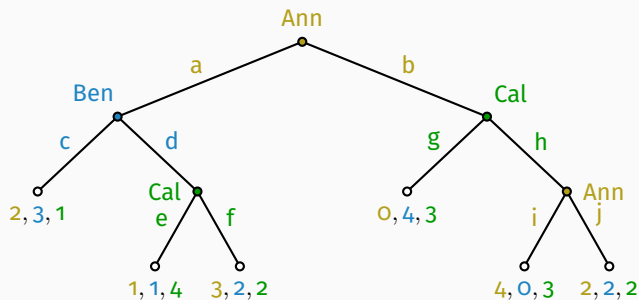
- The **payoffs** are wrong. Players may care about the other person or about fairness, so the game isn't the one we wrote down
- The **reasoning** is wrong. Players may not carry the unraveling through that many steps
- The **beliefs** are wrong. I might reason perfectly and still pass if I doubt that *you* will

All three are live. We'll get the machinery to separate them later.

Let's Work One Out

Planning Spring Break

Ann, Ben, and Cal share a house and settle the trip in turn, each seeing every decision made before theirs.



Payoffs in order: Ann, Ben, Cal.

Planning Spring Break

1. How many strategies does each player have?
2. Run backward induction. Where does a player face a tie?
3. Write out **both** backward-induction solutions, as strategy profiles.
4. What are the two backward-induction outcomes?

What Have We Learned, Charlie Brown?

- Perfect-information games are drawn as **rooted trees**; a frame becomes a game once preferences are added
- **Backward induction** works from the leaves inward; ties give several solutions
- A **strategy** covers all your decision nodes, even unreachable ones
- Every backward-induction solution is a **Nash equilibrium**; the extra Nash equilibria typically rest on **incredible threats**
- In finite two-player win-lose games, **somebody** has a winning strategy

Next: what happens when players *don't* see the whole history.

Reading & Practice

- Bonanno, *Game Theory*, Chapter 3
- Exercises: §3.7.1 (trees and frames), §3.7.2 (backward induction), §3.7.3 (strategies), §3.7.4 (two-player games)
- Exercise 3.8 shows IDWDS missing a backward-induction solution
- Solutions in §3.8